

Real Estate Investment Advisor: Property Price & Investment Trend Analysis

By Jiya Gupta

1. INTRODUCTION

The real estate industry generates large amounts of property-related data involving pricing, locations, property types, amenities, and investment trends. Analyzing this data helps buyers, investors, and businesses make informed decisions.

This project focuses on analyzing real estate property data using Python-based data analytics techniques. The project includes Exploratory Data Analysis (EDA), data visualization, and the development of an interactive Streamlit dashboard.

The dashboard allows users to explore property trends dynamically using filters and charts.

2. PROBLEM STATEMENT

Real estate buyers and investors often face difficulties in understanding property pricing patterns and identifying investment opportunities across different locations.

The objective of this project is to analyze property data and generate insights related to:

- Property pricing trends
- Investment opportunities
- Location-based pricing analysis
- Property features and amenities
- User-interactive dashboard visualization

3. OBJECTIVES

- Perform data cleaning and pre-processing.
- Conduct Exploratory Data Analysis (EDA).
- Identify key factors affecting property prices.
- Build interactive visualizations.
- Develop a Streamlit dashboard with user filters.
- Generate business and investment insights.

4. TOOLS & TECHNOLOGIES USED

Tool/Technology	Purpose
Python	Data Analysis
Pandas	Data Manipulation
NumPy	Numerical Operations
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
Plotly	Interactive Charts
Streamlit	Dashboard Development
VS Code	Development Environment
GitHub	Project Hosting
Streamlit Cloud	Deployment

5. DATA DESCRIPTION

The dataset contains information related to real estate properties across different cities and states.

Important Features

Feature	Description
State	State of property
City	City of property
Locality	Property locality
Property_Type	Apartment, Villa, House etc.
BHK	Number of bedrooms
Size_in_SqFtSize_in_SqFt	Property area
Price_in_Lakhs	Property price
Year_Built	Construction year
Furnished_Status	Furnishing condition
Nearby_Schools	Schools nearby
Nearby_Hospitals	Hospitals nearby
Parking_Space	Parking availability
Amenities	Amenities available
Public_Transport_Accessibility	Transport connectivity

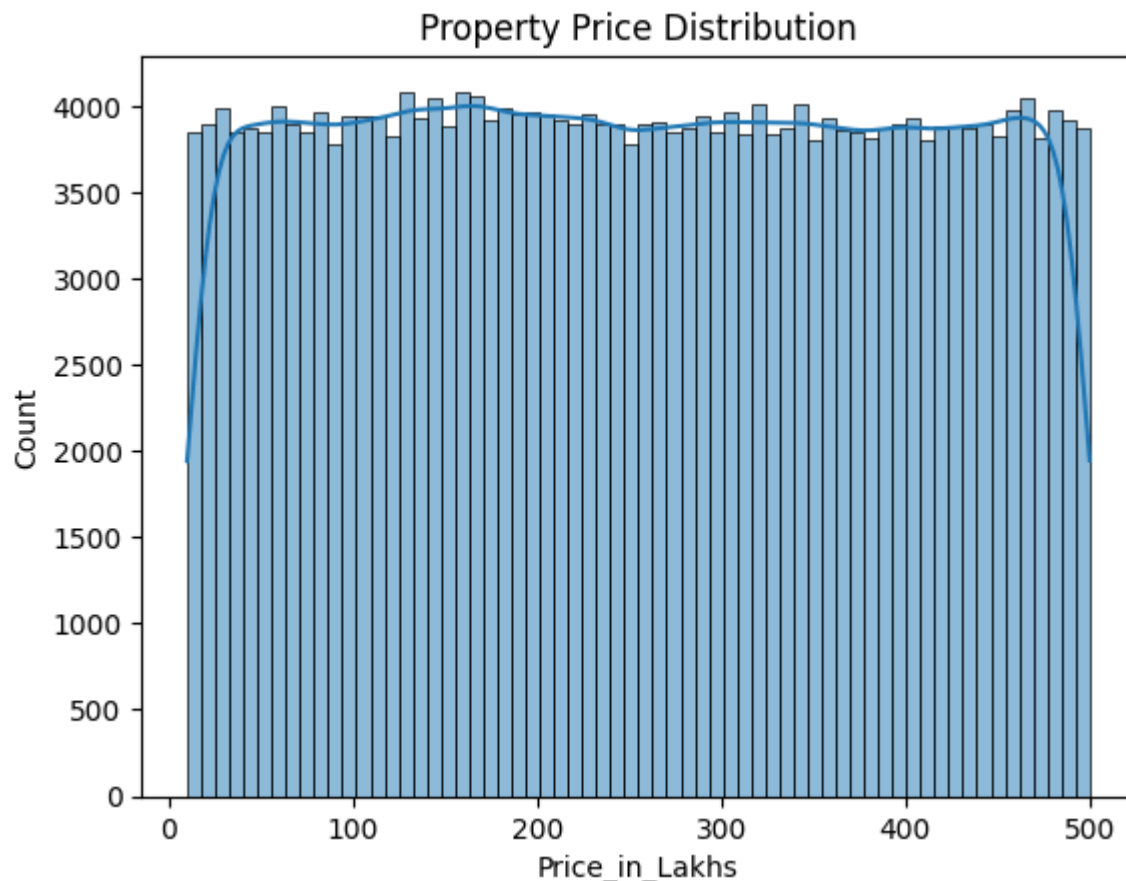
6. DATA CLEANING AND PRE-PROCESSING

- The following pre-processing steps were performed:
- Removed duplicate records.
- Handled missing values using forward fill method.
- Checked data types and dataset structure.
- Created additional features:
 - Age_of_Property
 - Price_per_SqFt
- Prepared cleaned dataset for analysis.

7. EXPLORATORY DATA ANALYSIS (EDA)

7.1 Price & Size Analysis

1. Distribution of Property Prices

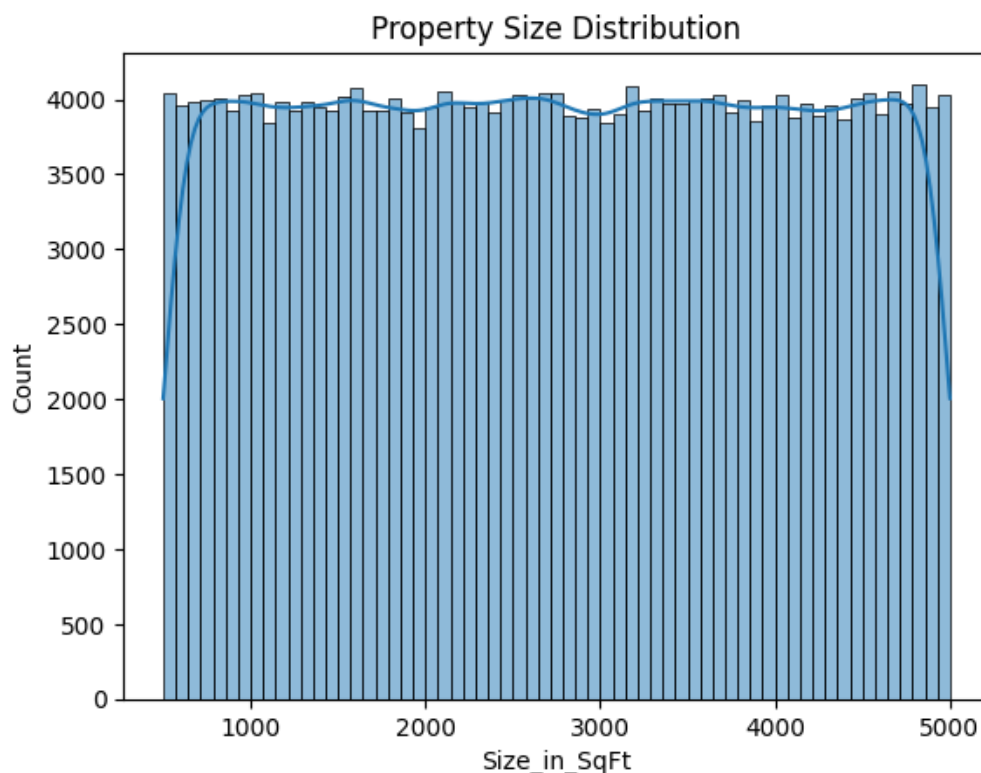


A histogram was used to analyze the distribution of property prices.

Insight

- Property prices are positively skewed.
- Luxury properties create high-value outliers.
- Most properties belong to medium price ranges

2. Distribution of Property Sizes

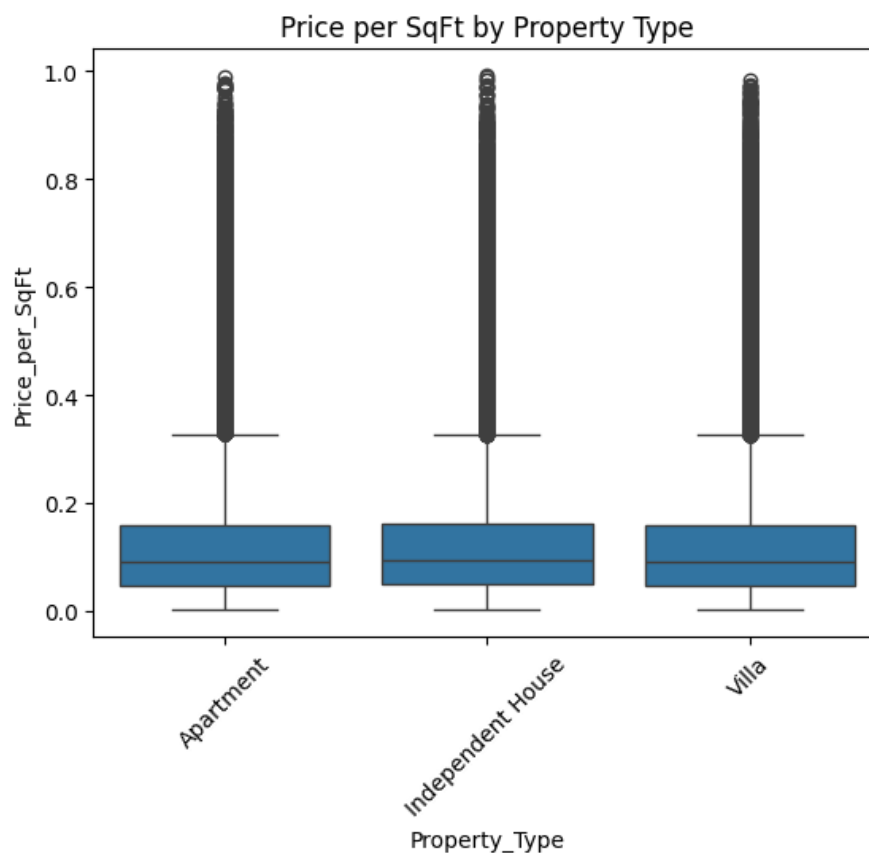


Property size distribution was analyzed using histogram visualization.

Insight

- Medium-sized properties dominate the dataset.
- Very large properties are comparatively fewer.

3. Price per SqFt by Property Type

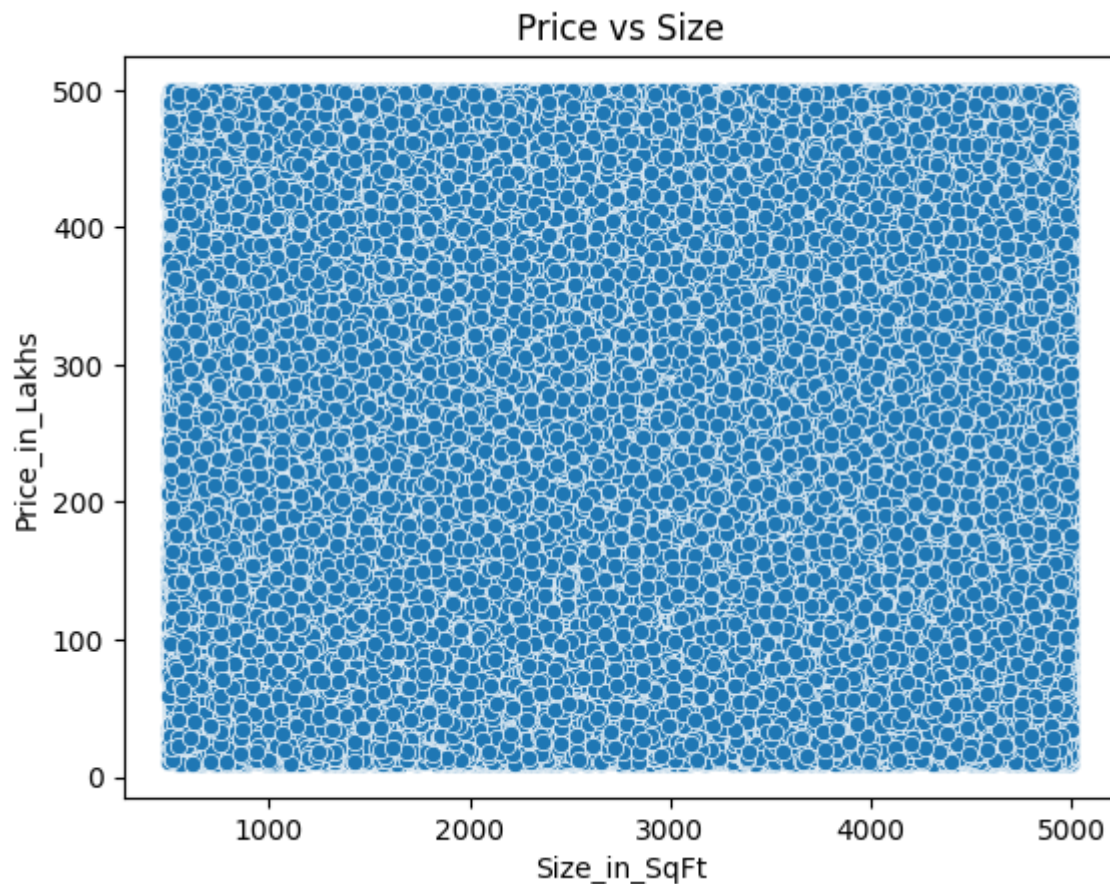


A boxplot was used to compare pricing across different property types.

Insight

- Villas and premium apartments have higher price per square foot.
- Property type strongly affects pricing.

4. Relationship Between Property Size and Price

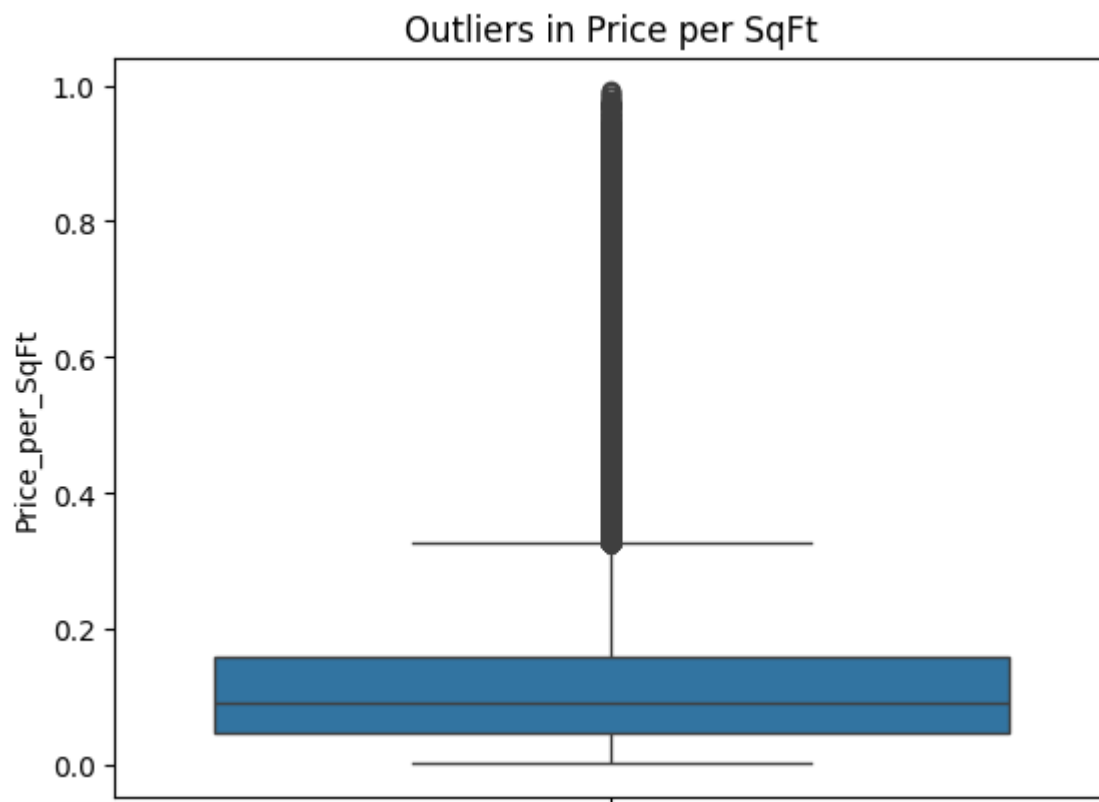


Scatter plots were used to study the relationship between property size and pricing.

Insight

- Larger properties generally have higher prices.
- Positive correlation exists between size and property value.

5. Outlier Detection in Pricing and Property Size



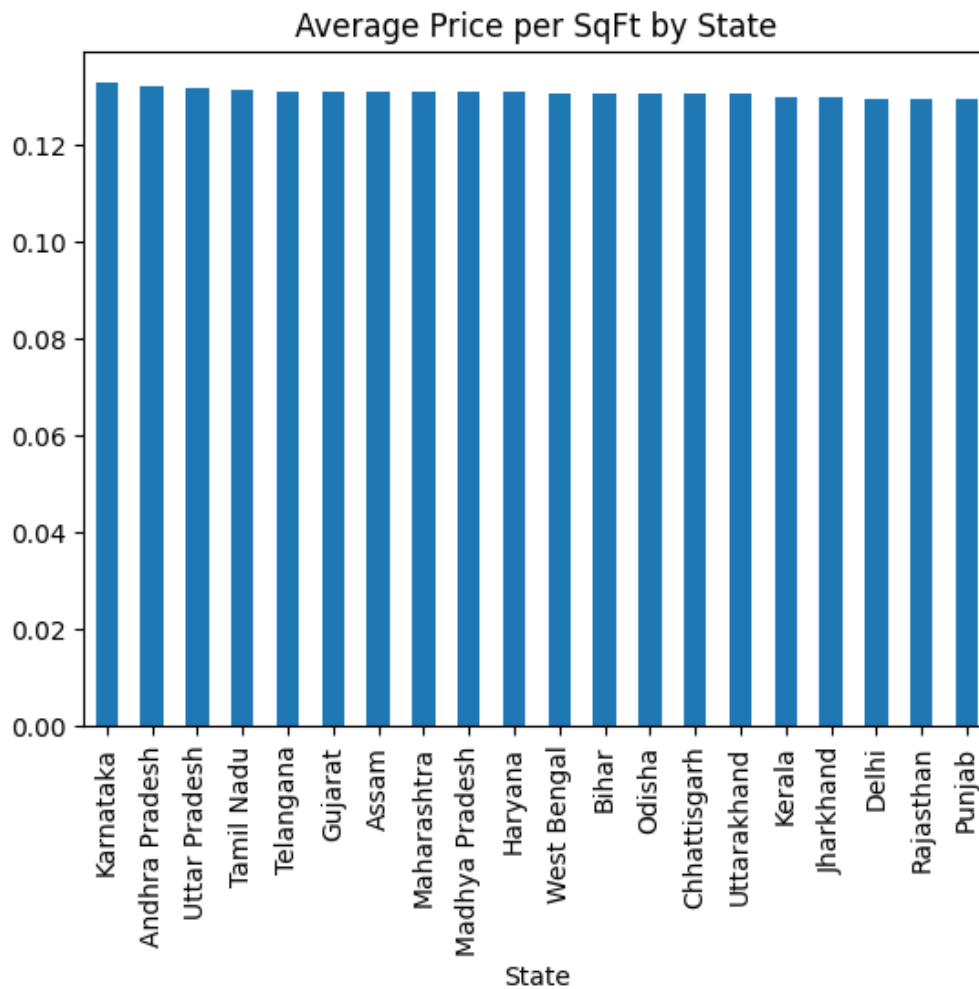
Boxplots were used to detect pricing and size outliers.

Insight

- Premium luxury properties act as outliers.
- High-end locations show extreme price values.

7.2 Location-Based Analysis

6. Average Price per SqFt by State

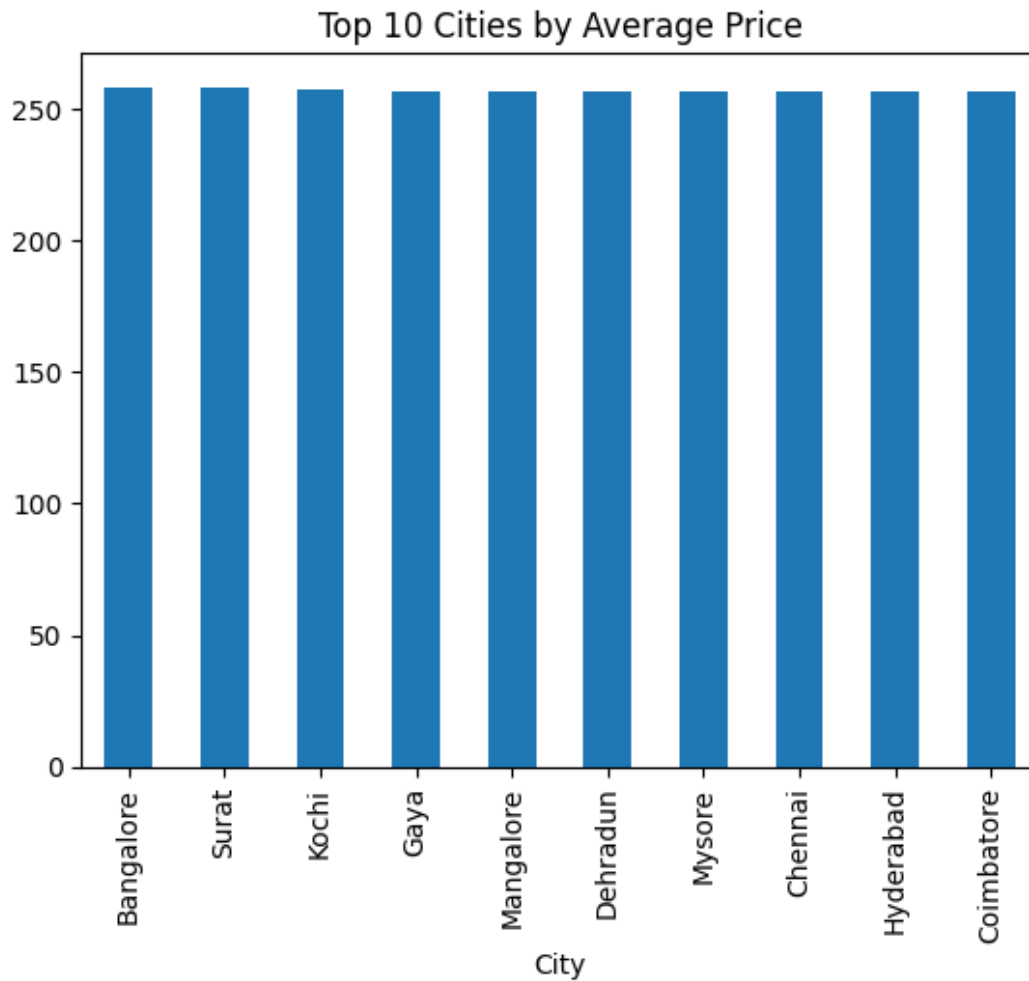


Bar charts were used for state-wise comparison.

Insight

- Metro and developed states have higher property values.
- Regional pricing differences are significant.

7. Average Property Price by City

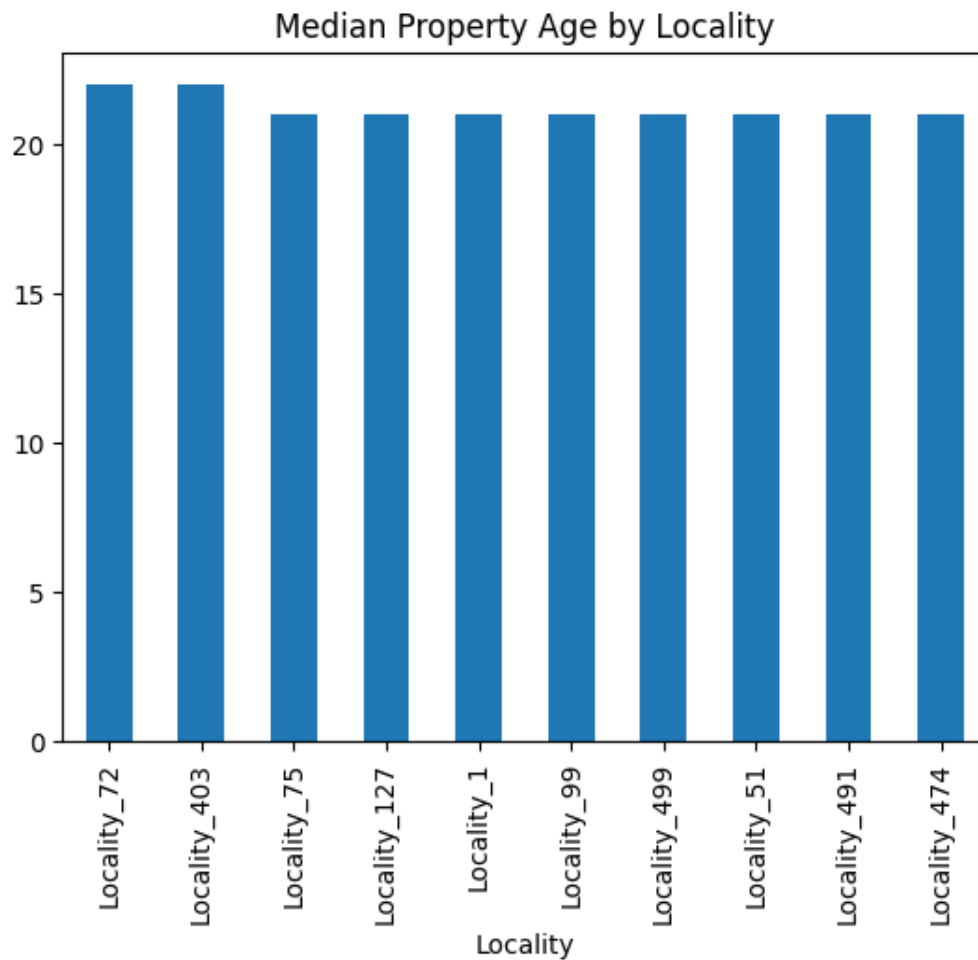


City-wise average pricing trends were analyzed.

Insight

- Metro cities dominate in average pricing.
- Urban demand increases property value

8. Median Age of Properties by Locality

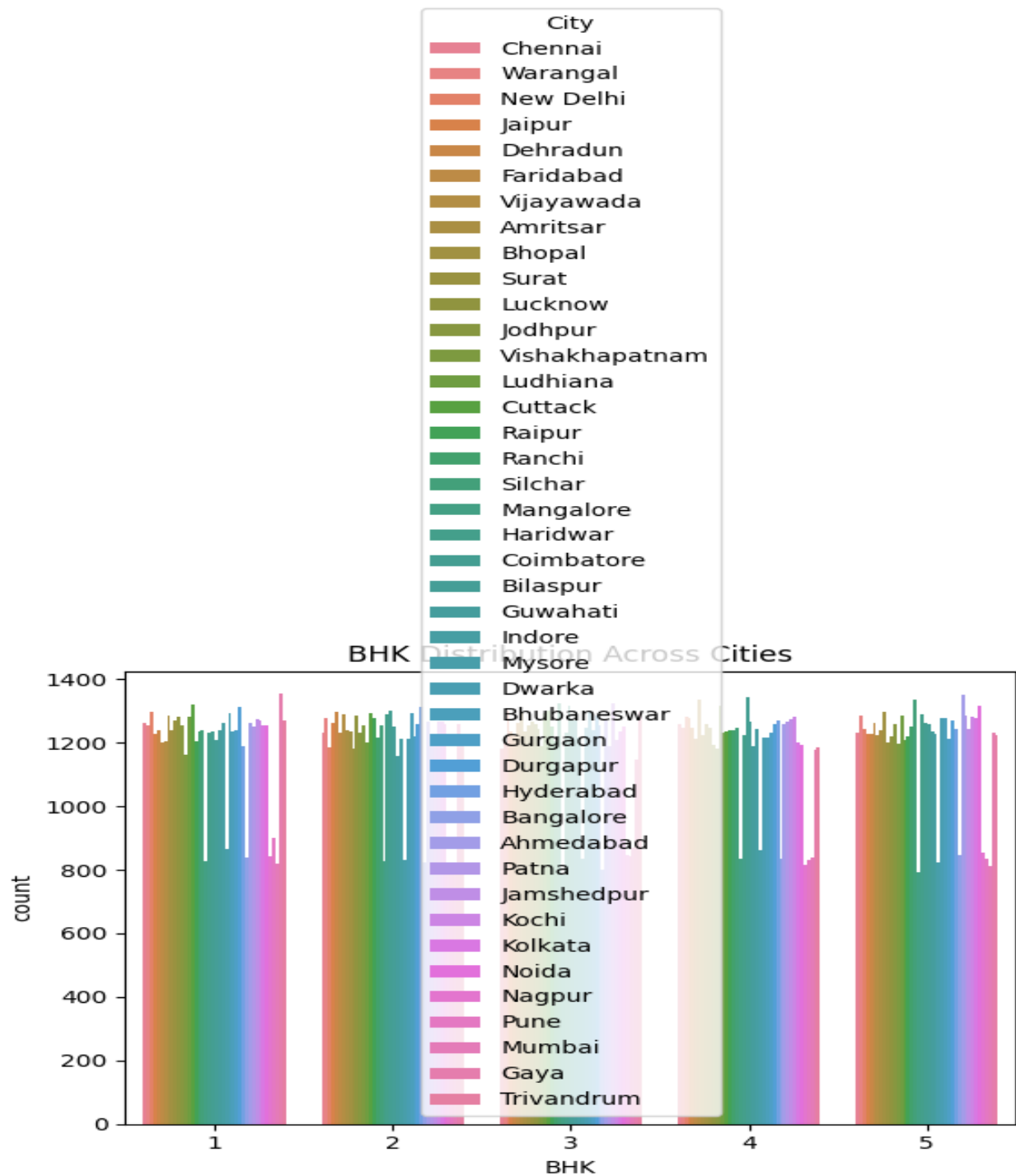


Median property age was analyzed locality-wise.

Insight

- Older localities contain aged properties.
- Newly developed areas contain modern projects

9. BHK Distribution Across Cities



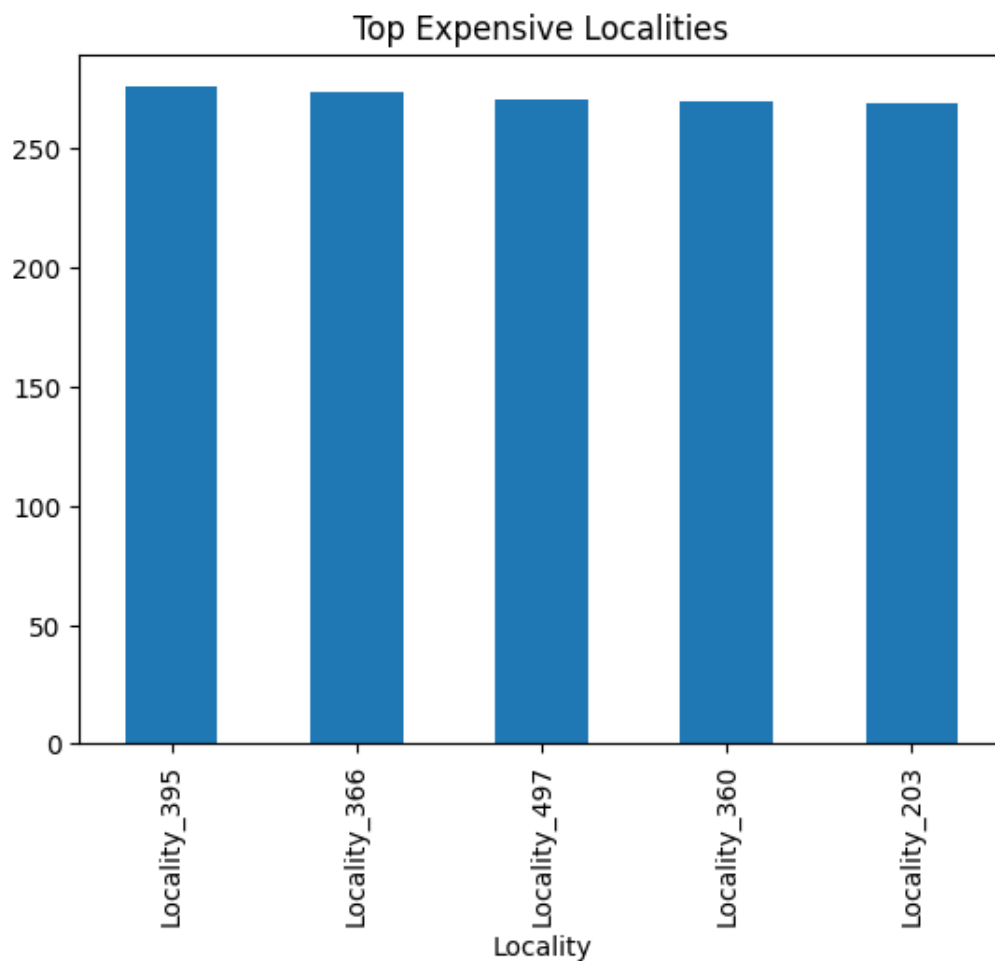
BHK distribution was analyzed using count plots.

Insight

- 2 BHK and 3 BHK properties dominate urban housing markets.

- Family-sized properties are most preferred.

10. Top Expensive Localities



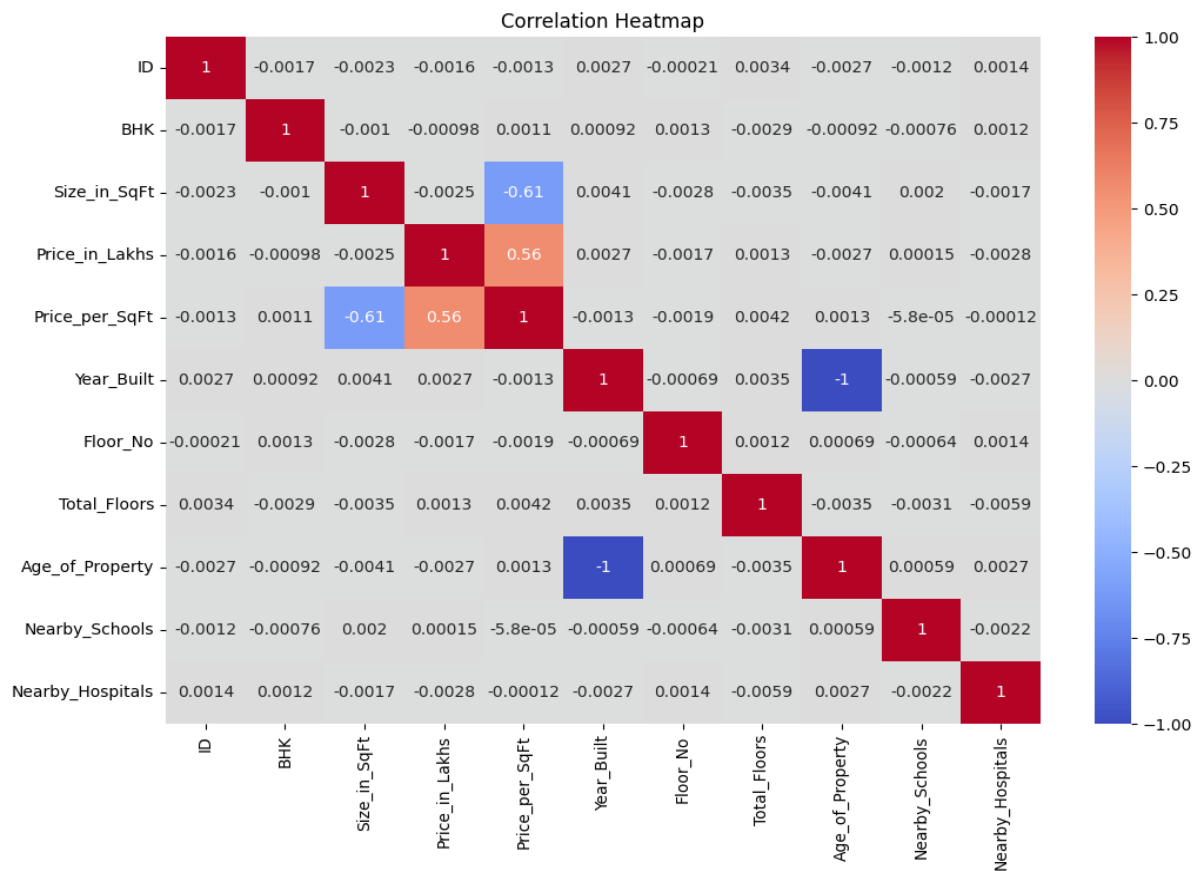
Top expensive localities were identified using average price analysis.

Insight

- Premium locations maintain consistently high prices.
- Infrastructure and connectivity increase property demand

7.3 Feature Relationship & Correlation Analysis

11. Correlation Between Numeric Features

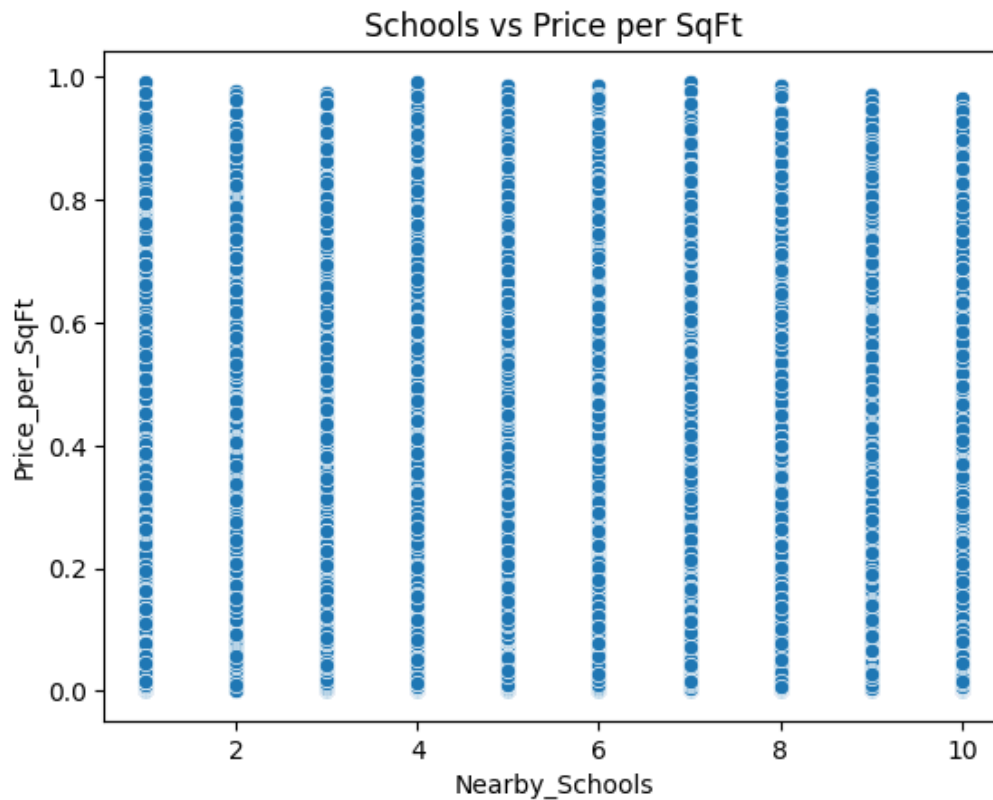


A heatmap was used to analyze relationships among numeric variables.

Insight

- Property size strongly correlates with property price.
- Nearby facilities also influence pricing.

12. Nearby Schools vs Price per SqFt

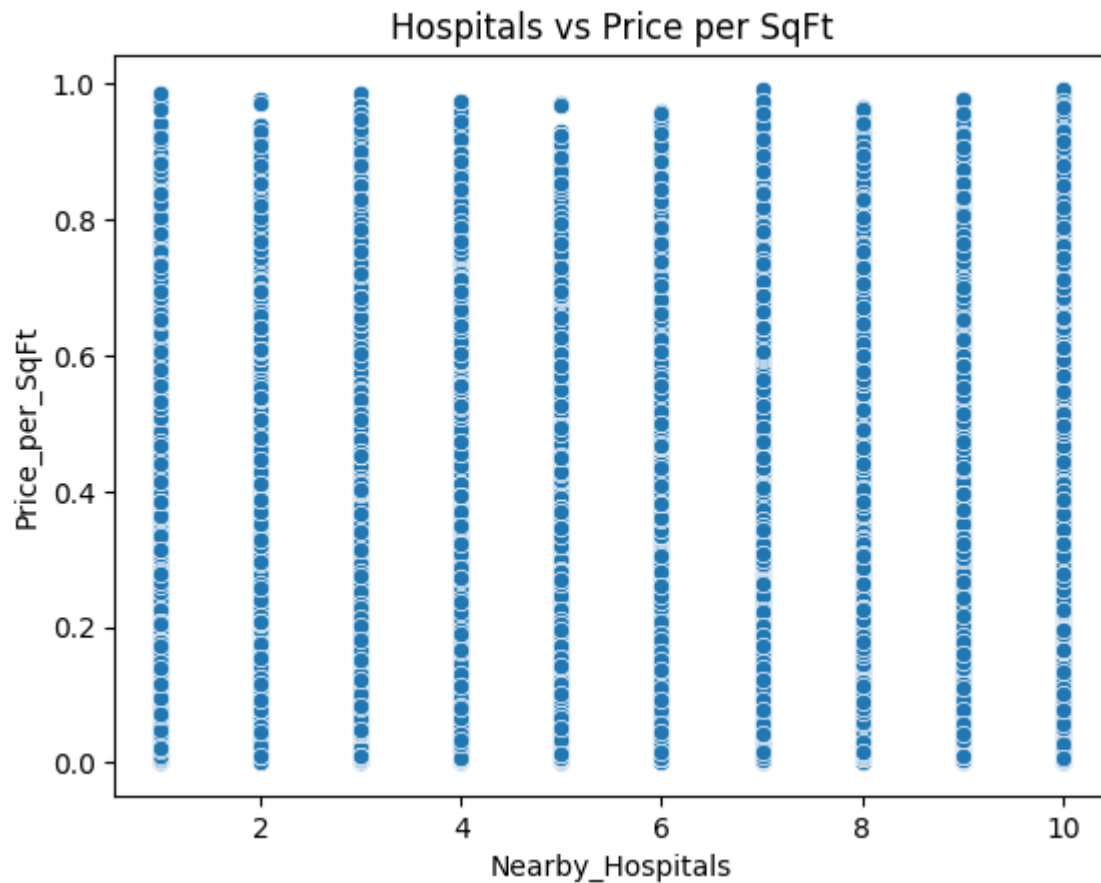


Scatter plots analyzed the impact of school accessibility.

Insight

- Properties near schools generally have higher prices.
- Educational infrastructure increases investment potential.

13. Nearby Hospitals vs Price per SqFt

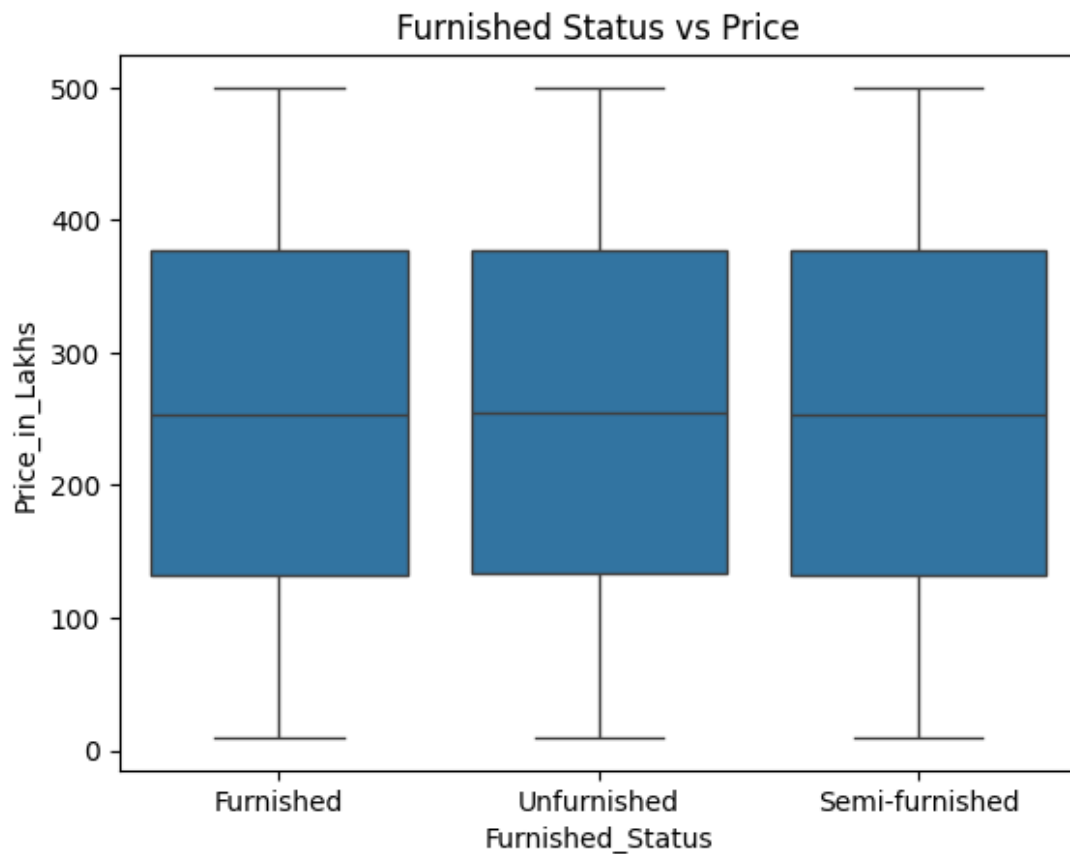


Hospital accessibility impact was analyzed.

Insight

- Better healthcare access positively affects property value.
- Residential demand increases near healthcare facilities.

14. Price Variation by Furnished Status

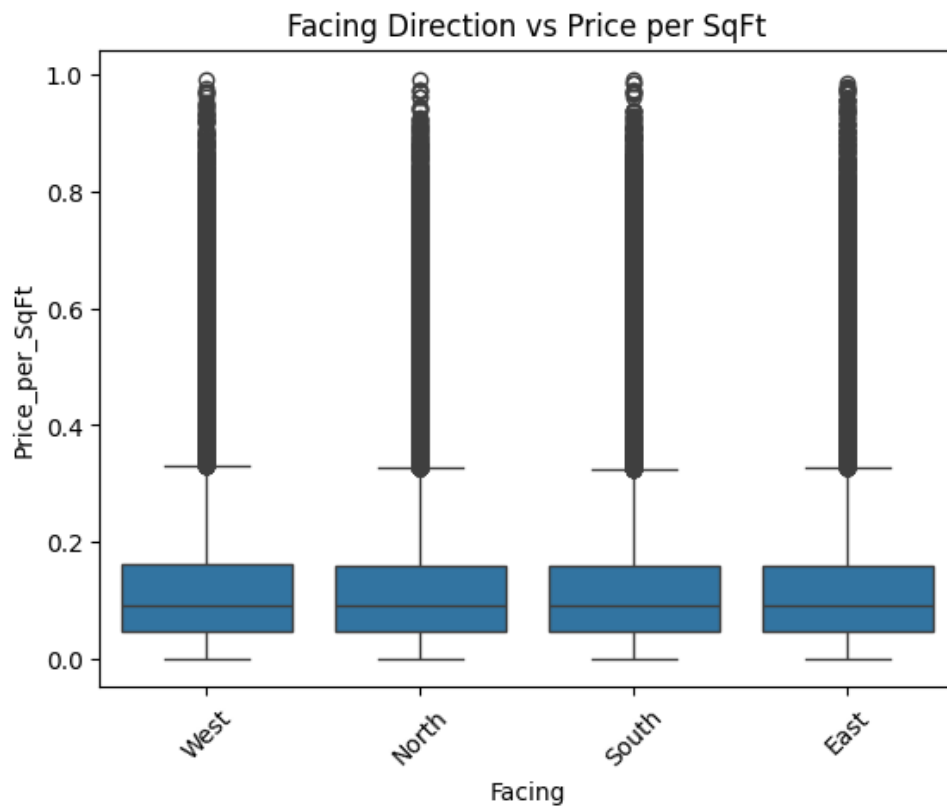


Boxplots compared property prices based on furnishing status.

Insight

- Furnished properties generally have higher prices.
- Furnishing adds premium value.

15. Facing Direction vs Price per SqFt



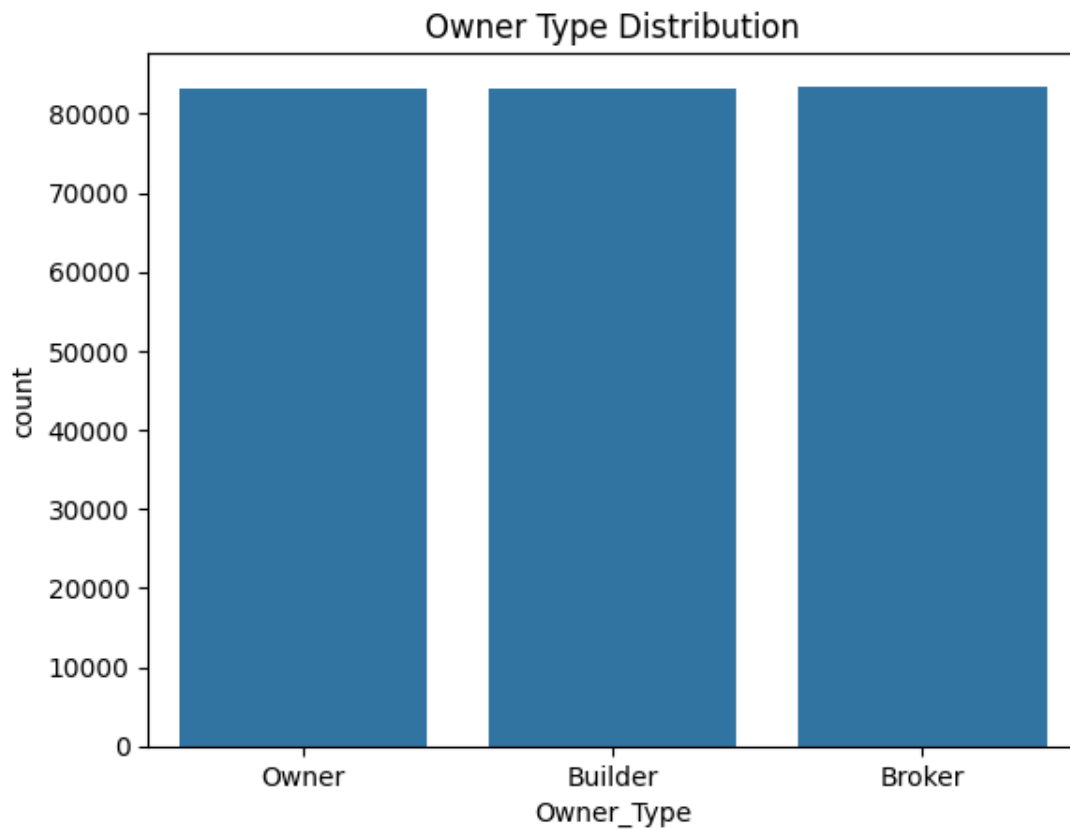
Facing direction impact on pricing was studied.

Insight

- Certain facing directions show slightly higher price trends.
- Buyer preferences influence pricing.

7.4 Investment & Amenities Analysis

16. Properties by Owner Type

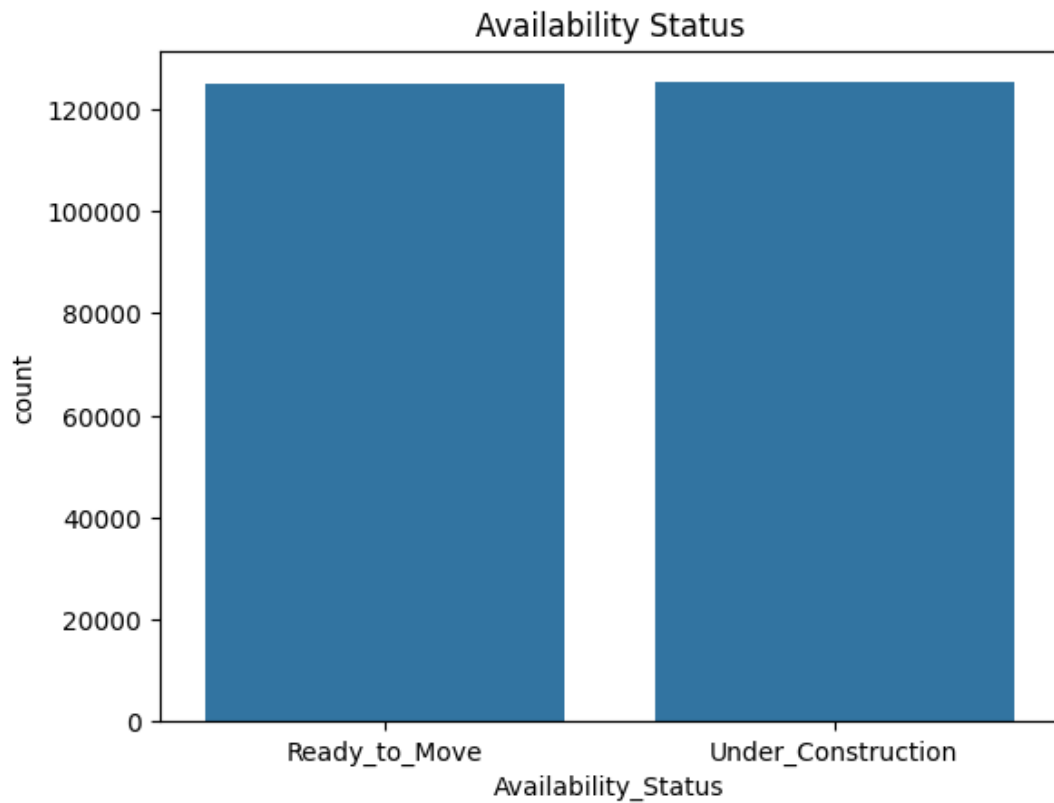


Owner type distribution was analyzed.

Insight

- Individual owners dominate the market.
- Builder properties are fewer but premium priced

17. Properties by Availability Status



Availability trends were studied.

Insight

- Most properties are available for purchase.
- Under-construction projects indicate future growth.

18. Parking Space vs Property Price

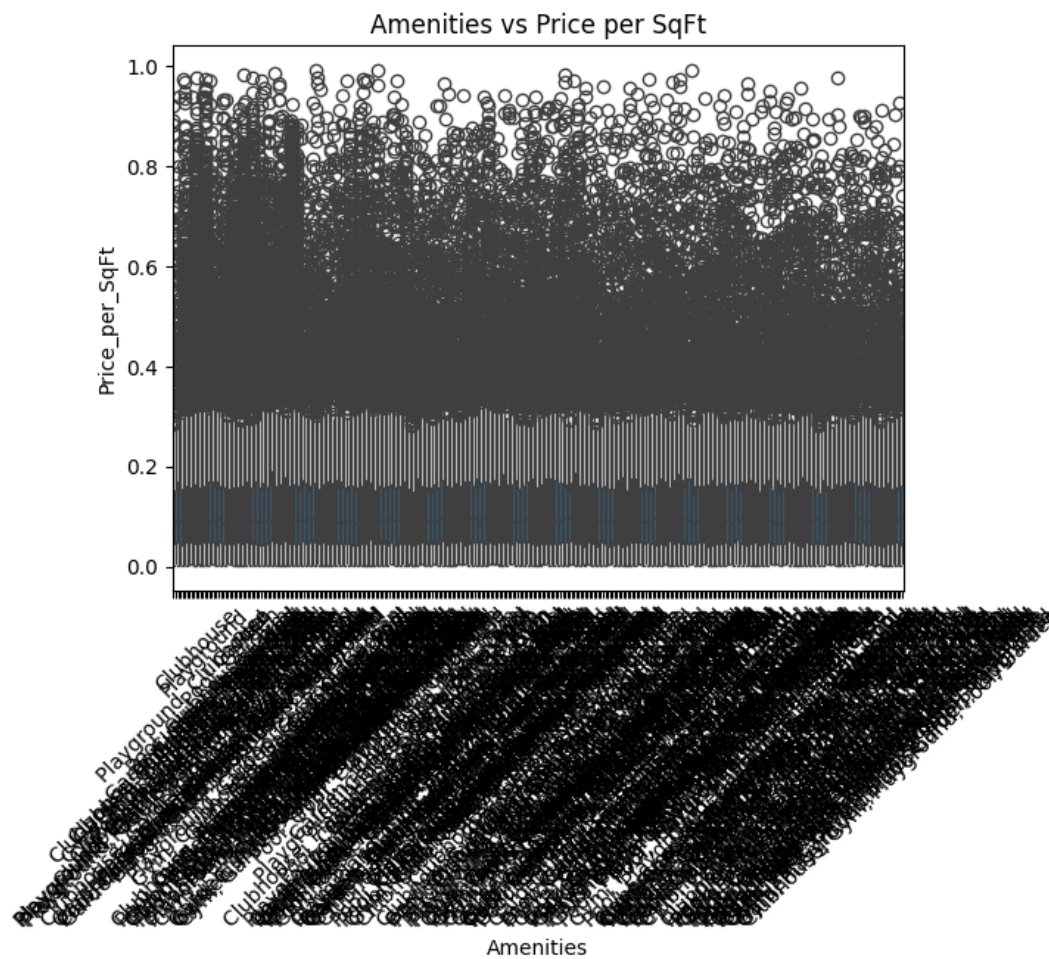


Parking availability impact on pricing was analyzed.

Insight

- Properties with parking spaces generally have higher prices.
- Parking is an important urban investment factor.

19. Amenities vs Price per SqFt

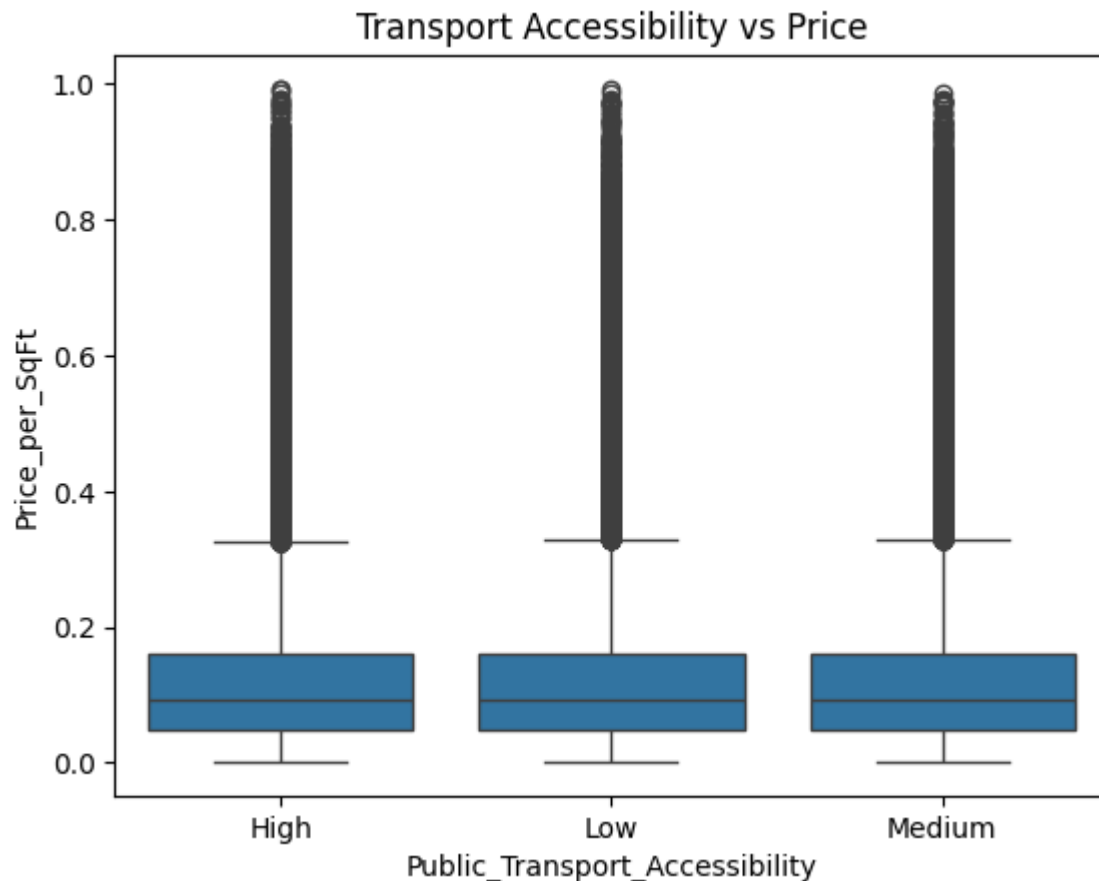


Amenities impact on pricing was studied.

Insight

- Luxury amenities increase property value.
- Better facilities improve customer demand.

20. Public Transport Accessibility vs Investment Potential



Transport accessibility analysis was performed.

Insight

- Better public transport connectivity improves investment attractiveness.
- Metro connectivity positively affects pricing.

8. STREAMLIT DASHBOARD

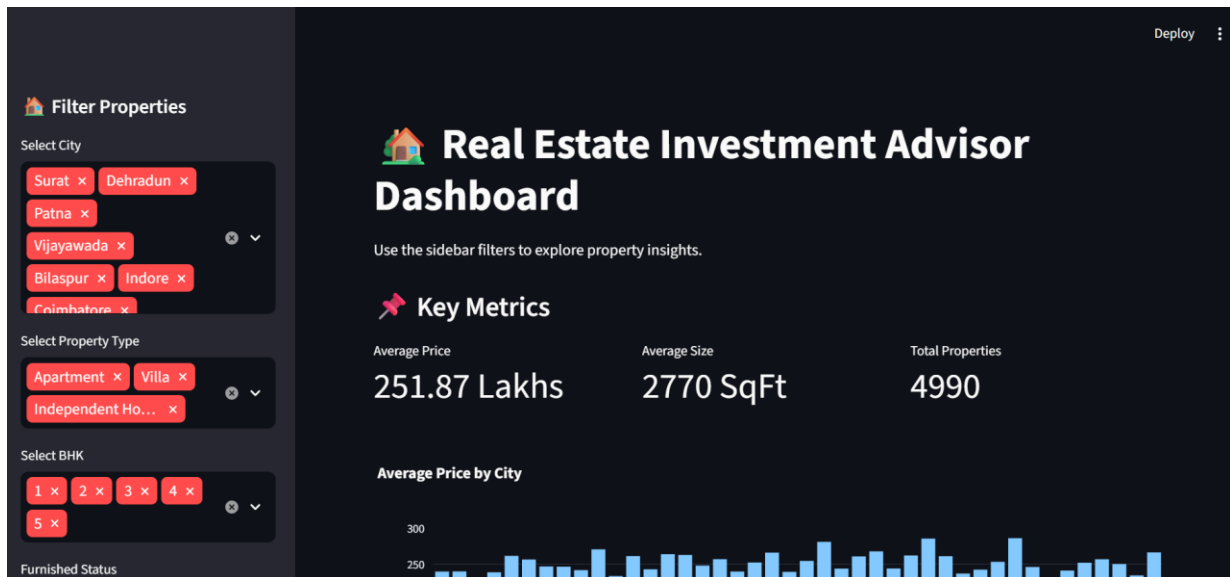
An interactive Streamlit dashboard was developed to visualize property insights dynamically.

Dashboard Features

- User Filters
- City
- Property Type
- BHK
- Furnished Status
- Price Range

KPI Cards

- Average Price
 - Average Property Size
 - Total Properties
-
- Interactive Charts
 - Average Price by City
 - Price vs Property Size
 - Furnished Status vs Price
 - Property Type Distribution
 - Price per SqFt by State
 - BHK Distribution
 - Owner Type Distribution
 - Parking Space vs Price



9. PROJECT DEPLOYMENT

GitHub Repository Link

<https://github.com/Jiya1001/real-estate-dashboard/blob/main/app.py>

Streamlit Dashboard Link

<https://real-estate-dashboard-6ixqqldcqw3wekqvjvph2o.streamlit.app/>

10. KEY BUSINESS INSIGHTS

- Metro cities have significantly higher property prices.
- Furnished properties attract premium pricing.
- Properties near schools and hospitals show better investment potential.
- Public transport accessibility increases property demand.
- Amenities and parking availability positively affect property value.

- 2 BHK and 3 BHK properties dominate urban markets.
- Modern and newly developed properties attract buyers and investors.

11. CONCLUSION

The project successfully analyzed real estate property data and generated meaningful insights related to pricing trends, investment opportunities, and customer preferences.

The interactive Streamlit dashboard allows users to dynamically explore property data using filters and visualizations.

The project demonstrates how data analytics and visualization techniques can support better real estate investment decisions.